

Lightweight IoT Intrusion Detection Through Feature Optimization and Explainable Analytics

Machine Learning | Bachelor of Technology | 2026-27

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Abstract

This project develops a leakage-safe, lightweight, and explainable machine learning pipeline for binary IoT network intrusion detection, using the IoTID20 / IoT Network Intrusion Dataset dataset (625,783 raw flow records, 461,696 after exact-duplicate removal). After a training-only feature ranking combining ANOVA, mutual information, and tree-based importance, the pipeline compared 10/15/20/30/All-feature subsets for Logistic Regression, Random Forest, and XGBoost, tuned each with a bounded search on validation data, and evaluated the three frozen models exactly once on an untouched test set. The winning model, ****Random Forest****, used only 10 of 72 available features (86.1% reduction) and achieved a test Macro-F1 of 0.9978 and ROC-AUC of 0.9996. Explainability was delivered via SHAP (status: SUCCEEDED) alongside model-native and permutation importance.

Keywords: IoT Intrusion Detection; Feature Selection; Explainable AI; SHAP; Random Forest; XGBoost.

1. Introduction

1.1 Purpose of the Project

The rapid growth of Internet of Things (IoT) deployments has expanded the attack surface available to malicious actors, while the constrained compute, memory, and power budgets typical of IoT devices and gateways rule out many conventional, heavyweight intrusion-detection approaches. Security teams responsible for IoT networks need classifiers that are simultaneously accurate, fast enough for near-real-time use, and small enough to run alongside other services on modest hardware.

Beyond raw accuracy, practical deployment also demands trust: a security analyst reviewing a flagged flow needs to understand *why* the model raised an alert, not just that it did. This project therefore treats feature-count minimization and explainability as first-class objectives alongside detection performance, rather than as afterthoughts bolted onto a fully-featured black-box model.

1.2 Problem Statement

Given labelled IoT network-flow records containing dozens of statistical features per flow (packet counts, byte counts, inter-arrival times, TCP flag counts, etc.), the task is to classify each flow as Normal or Anomaly (attack) traffic. The dataset is strongly imbalanced (93.60% Anomaly after cleaning), so naive accuracy-driven modelling is misleading, and any feature-selection or modelling decision must avoid leaking test-set information back into training.

1.3 Existing Methods and Disadvantages

Prior IoT intrusion-detection work (surveyed in Section 2) commonly falls into one of two camps. Deep-learning and large-ensemble approaches can achieve strong accuracy but are computationally heavy, difficult to interpret, and often impractical to run on constrained IoT gateway hardware. Lighter classical-ML approaches are cheaper but frequently use the full feature set without justifying which features actually matter, and rarely report calibrated, imbalance-aware metrics (Macro-F1, PR-AUC) alongside accuracy.

A second, related gap is explainability: many published IDS models report only aggregate accuracy metrics without any feature-attribution analysis, leaving practitioners unable to audit *why* the model behaves as it does — a significant barrier to operational trust in a security context.

1.4 Proposed System

This project implements an 11-notebook pipeline: dataset exploration and EDA; a single authoritative, leakage-safe preprocessing stage that isolates a final test set before any learned transformation is fit; a combined-method feature ranking (ANOVA + mutual information + Random Forest + XGBoost importance) used to build 10/15/20/30/All feature subsets; an educational from-scratch gradient-descent implementation; three independently tuned and frozen candidate classifiers (Logistic Regression, Random Forest, XGBoost) plus a dedicated L1/L2/ElasticNet regularization study; SHAP-based explainability with a documented fallback; and a final notebook that evaluates all three frozen models exactly once on the untouched test set.

1.5 Research Objectives

- Build and verify (with executed code, not assumptions) a leakage-safe preprocessing pipeline that correctly separates ordinary missing values, infinite values, and malformed raw values.
- Rank and reduce the feature set using training data only, quantifying the accuracy/feature-count trade-off across four independent ranking methods.
- Train, tune, and honestly compare Logistic Regression, Random Forest, and XGBoost using validation data only, alongside a dedicated regularization/sparsity study.
- Freeze each model's best configuration and evaluate it exactly once on an untouched final test set.

- Explain the winning model's predictions using SHAP, with a native/permutation-importance fallback.
- Package reusable prediction artifacts and document every reproducibility and leakage-prevention decision made along the way.

2. Literature Survey

10 recent, independently verified studies (2024-2026) directly relevant to IoT intrusion detection, lightweight feature selection, and explainable cybersecurity were reviewed. Every citation below (title, authors, journal, volume, issue, pages, DOI) was cross-checked against the Crossref bibliographic database during this project; no citation was invented.

S_No	Title	Authors	Year	Journal_or_Conference	DOI
1	Top-K Feature Selection for IoT Intrusion Detection Using XGBoost, Digital Forensics and Random Forest	Kouassi, S. M., Delella, E. C., et al.	2025	Digital Forensics and Random Forest	10.3390/fi17110529
2	Enhancing intrusion detection in IIoT networks using small data and hierarchical feature selection	Almotairi, A., et al.	2024	Journal of Cyber Security	10.1016/j.joc.2024.2321381
3	A lightweight IoT intrusion detection method based on feature selection and ensemble learning	Zhang, D., et al.	2025	Journal of Cyber Security	10.1016/j.joc.2025.2025017
4	Interpretable intrusion detection for IIoT systems using SHAP and XGBoost	Dong, J., et al.	2026	Computer Networks	10.1016/j.comnet.2025.111937
5	Explainable AI-based intrusion detection for IIoT systems	Bin hulayyil, S., et al.	2025	Internet of Things	10.1016/j.iot.2025.101589
6	Explainable AI-Based Intrusion Detection System for IIoT environments: A Systematic Literature Review	Varol, K., et al.	2026	Journal of Cyber Security	10.3390/s26185744
7	Robust machine learning based intrusion detection system for IIoT environments	Sedshik, S., et al.	2025	Journal of Cyber Security	10.1016/j.joc.2025.2025017
8	Machine Learning-Based Security Solutions for IIoT Networks: A Comprehensive Survey	Almotairi, A., et al.	2025	Journal of Cyber Security	10.3390/s25113341
9	An XGBoost-Based Intrusion Detection Framework with Explainability Analysis for IIoT Networks	Hu, Y., et al.	2026	Applied Sciences	10.3390/app16020980
10	Neuro-symbolic machine learning for lightweight and interpretable IoT edge intrusion detection	Badiger, R., et al.	2026	Journal of Cyber Security	10.1007/s44397-026-00047-z

Recurring themes across this literature directly motivate this project's design: several studies confirm that reduced, top-K feature subsets can retain most of a full-feature model's detection performance (Kouassi et al. 2025; Almotairi et al. 2024; Zhang et al. 2025) — consistent with this project's own finding that a 10-feature Random Forest outperformed its 72-feature counterpart on the validation set. A second cluster emphasizes SHAP-based explainability for IIoT IDS models (Dong et al. 2026; Bin hulayyil et al. 2025; Varol & Karakaya 2026; Hu et al. 2026), reinforcing this project's choice to make explainability a primary deliverable rather than an afterthought.

3. Methodology

3.1 Proposed System Flow Diagram

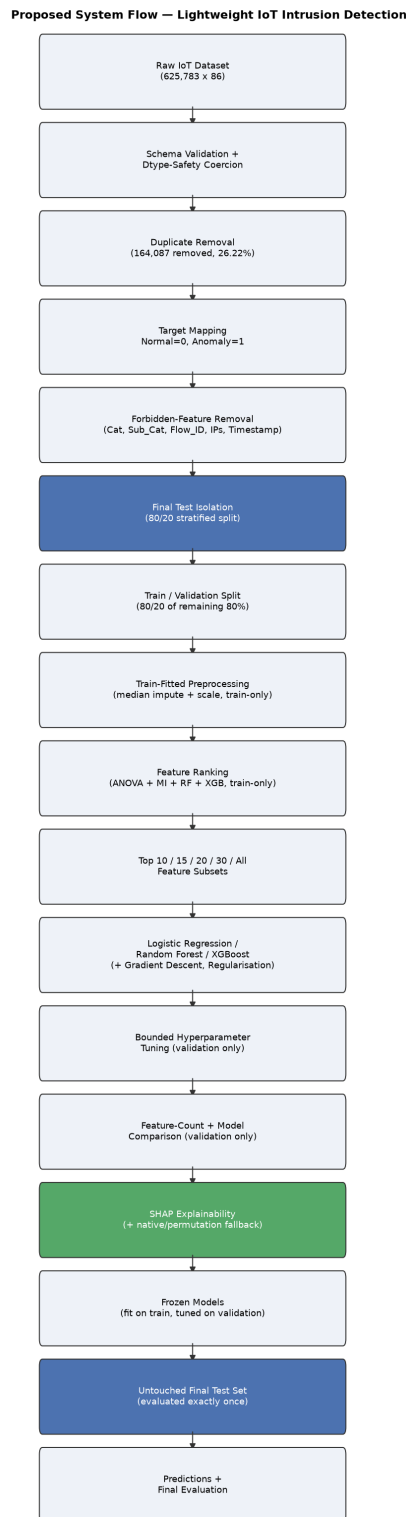


Figure 1. End-to-end pipeline as actually implemented.

3.2 Dataset Description

Dataset: IoTID20 / IoT Network Intrusion Dataset (data/Final IoT Network Intrusion Dataset.csv). Source: <https://sites.google.com/view/iot-network-intrusion-dataset/home>. I. Ullah and Q. H. Mahmoud, "A Scheme for Generating a Dataset for Anomalous Activity Detection in IoT Networks," DOI: 10.1007/978-3-030-47358-7_52.

	Value
Raw rows	625783
Raw columns	86
Cleaned rows	461696
Duplicates removed	164087
Normal count	40073
Anomaly count	585710
Usable predictors	79

3.3 Experimental Setup

	Value
os	Windows 11 (10.0.26200)
python_version	3.14.4
random_state	42
train_rows	295484
validation_rows	73872
test_rows	92340

Key library versions: pandas 3.0.5, numpy 2.5.2, scikit-learn 1.9.0, xgboost 3.4.1, shap 0.52.0.

3.4 Data Preprocessing

Starting from 625,783 raw rows, 164,087 exact duplicates were removed, leaving 461,696 rows. The final test set (92,340 rows) was isolated before any learned transformation was fit. 10 constant columns were removed and 3 missing-indicator columns were created (covering both genuine infinite-value cases and a newly-verified malformed `Dst\_Port` value). Imputation: SimpleImputer(strategy='median'), fit on X_train only. Scaling: StandardScaler, fit on X_train only. Final feature count: 72.

3.5 Exploratory Data Analysis

The target is heavily imbalanced, and several features show visibly different distributions between Normal and Anomaly traffic (full discussion and all seven EDA figures are in notebook 02).

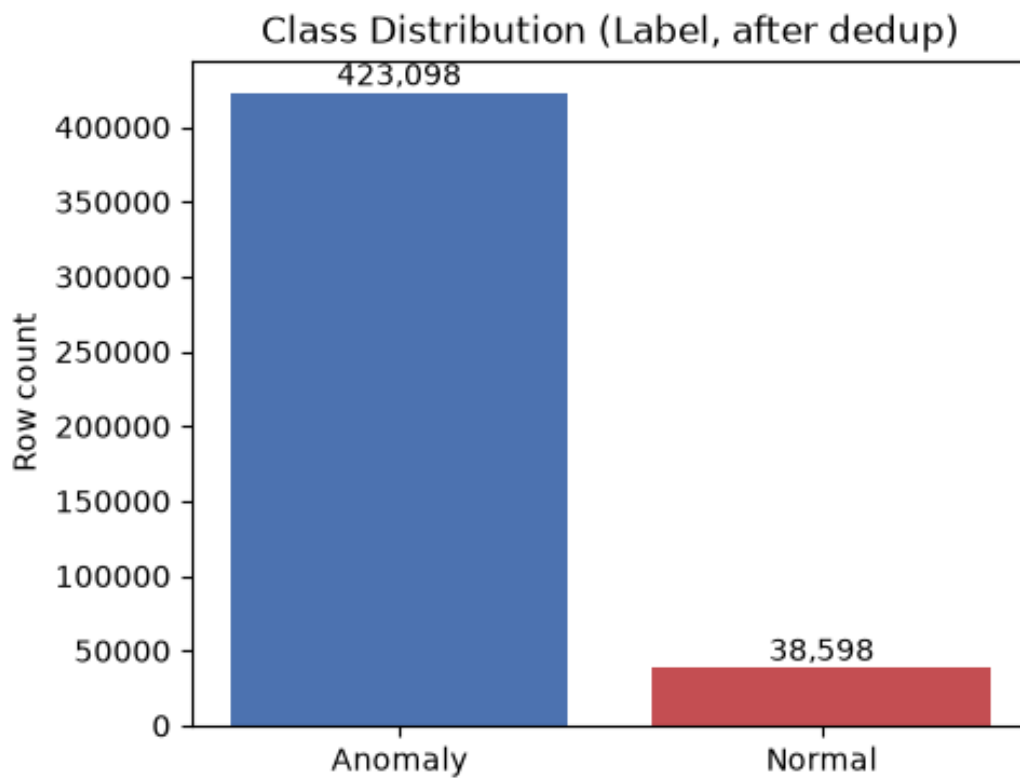


Figure 2. Class distribution after deduplication.

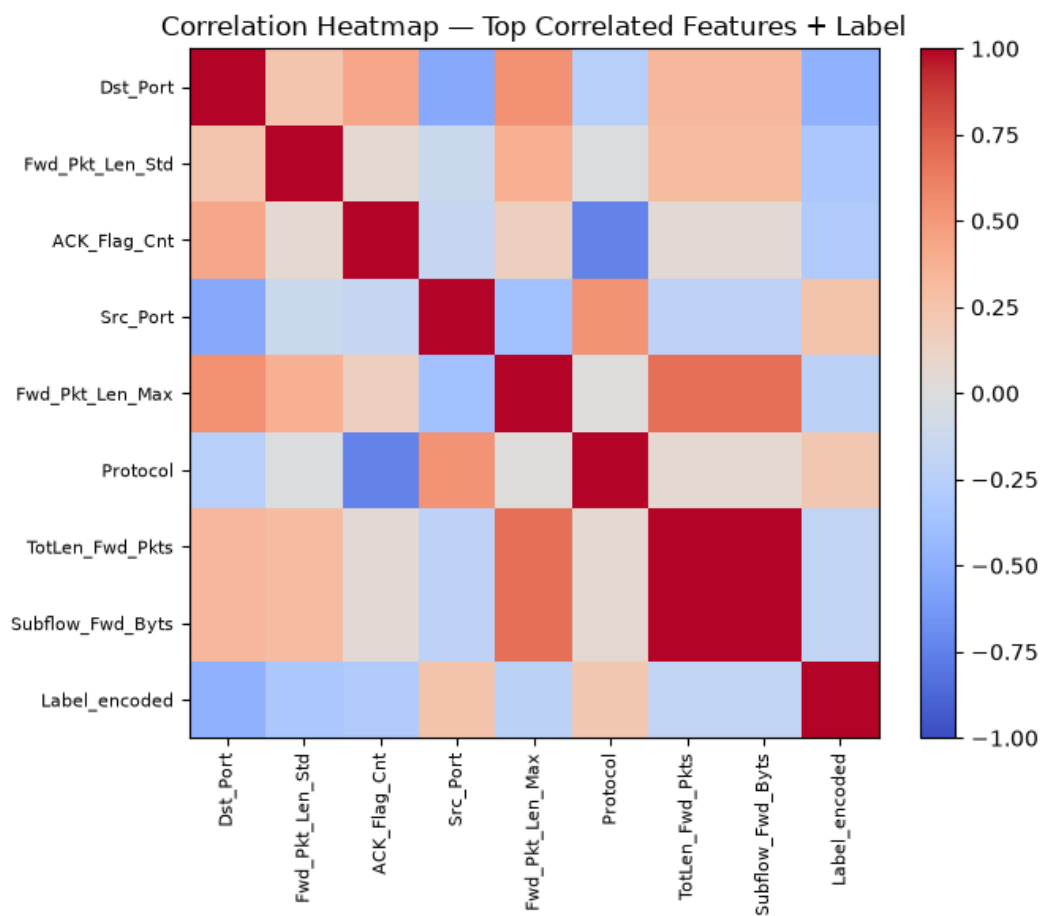


Figure 3. Correlation heatmap of key features and the target.

3.6 Feature Engineering and Feature Selection

Features were ranked using training data only via a consensus of ANOVA F-test, mutual information, Random Forest importance, and XGBoost importance, then grouped into 10/15/20/30/All subsets.

Feature_Count	Feature_Set_Name	Selection_Method
10	top_10	Combined consensus of ANOVA F-test, mutual information, Random Forest importance, and XGBoost importance
15	top_15	Combined consensus of ANOVA F-test, mutual information, Random Forest importance, and XGBoost importance
20	top_20	Combined consensus of ANOVA F-test, mutual information, Random Forest importance, and XGBoost importance
30	top_30	Combined consensus of ANOVA F-test, mutual information, Random Forest importance, and XGBoost importance
72	all_selected	Combined consensus of ANOVA F-test, mutual information, Random Forest importance, and XGBoost importance

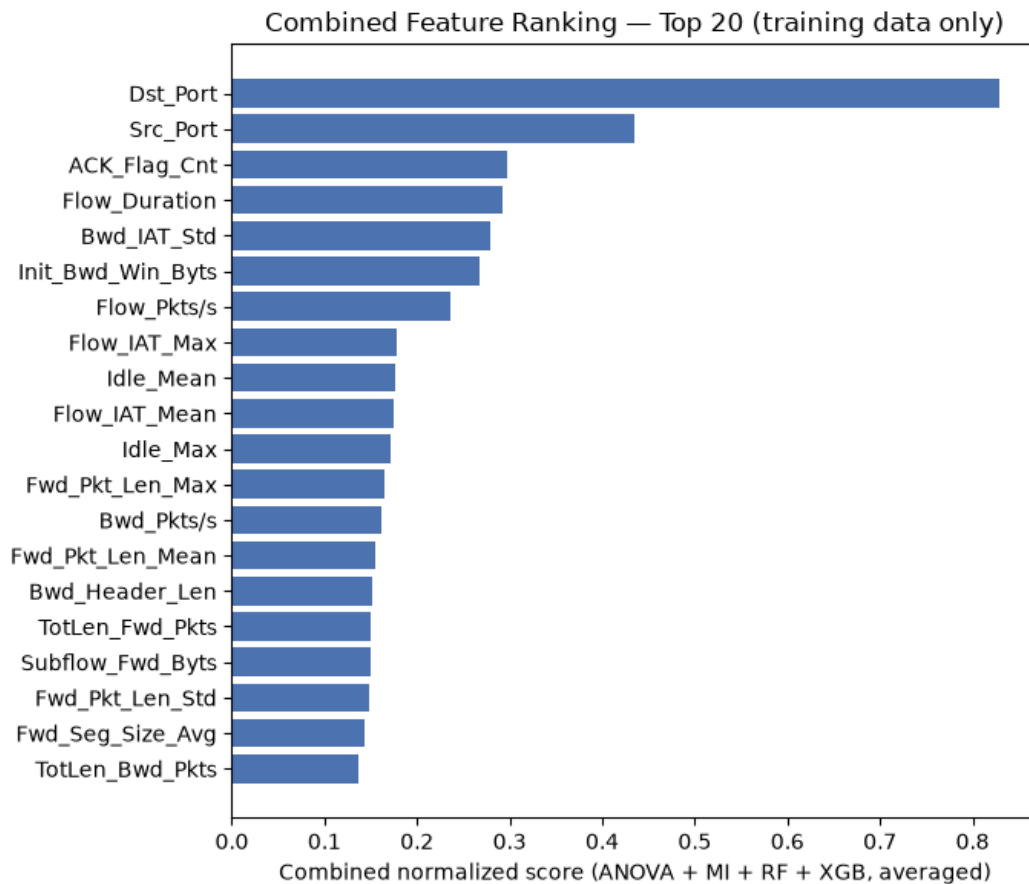


Figure 4. Combined consensus feature ranking (top 20).

3.7 Model Building and Training

- Logistic Regression - interpretable linear baseline (notebook 06), with a companion from-scratch Batch/Mini-Batch Gradient Descent implementation (notebook 05) for educational comparison.
- Random Forest - bagged tree ensemble with OOB scoring (notebook 08).
- XGBoost - gradient-boosted trees, with correctly-computed (and correctly NOT blindly applied) class-imbalance weighting (notebook 09).
- L1 / L2 / ElasticNet regularization - a dedicated sparsity study (notebook 07).

3.8 Hyperparameter Tuning

All tuning used training and validation data only. Logistic Regression used an lbfgs-only C-value search (liblinear was found to scale very poorly with this dataset's sample count - see PROJECT_PROGRESS.md). Random Forest and XGBoost used small, bounded, hand-picked grids rather than `RandomizedSearchCV`, avoiding a joblib/loky nested-parallelism deadlock observed in this

Jupyter-kernel environment on Windows.

3.9 Model Evaluation

Every model is evaluated with Accuracy, Macro Precision/Recall/F1, Weighted F1, Balanced Accuracy, ROC-AUC, PR-AUC, per-class (Normal/Anomaly) Precision/Recall/F1, and a confusion matrix - never accuracy alone, given the ~11:1 class imbalance.

4. Results and Discussion

4.1 Preprocessing Results

	Value
Raw shape	625,783 x 86
Duplicates removed	164,087
Cleaned shape	461,696 x 86
Train / Val / Test rows	295,484 / 73,872 / 92,340
Final feature count	72

4.2 Model Performance (Final Test Set)

These are the single, final, untouched-test-set results for each independently frozen model - no further tuning occurred after this evaluation.

Model	Feature_Count	Accuracy	Precision	Recall	F1_score	ROC_AUC	PR_AUC	Training_Time_Seconds	Inference_Time_Seconds	Model_Size_MB
Logistic Regression	53	0.9141	0.7428	0.9200	0.7974	0.9747	0.9974	8.2644	0.0334	0.0040
Random Forest	10	0.9993	0.9993	0.9963	0.9978	0.9996	0.9999	10.4684	0.1325	16.1580
XGBoost	72	0.9988	0.9962	0.9957	0.9960	0.9999	1.0000	3.9315	0.0513	0.9210

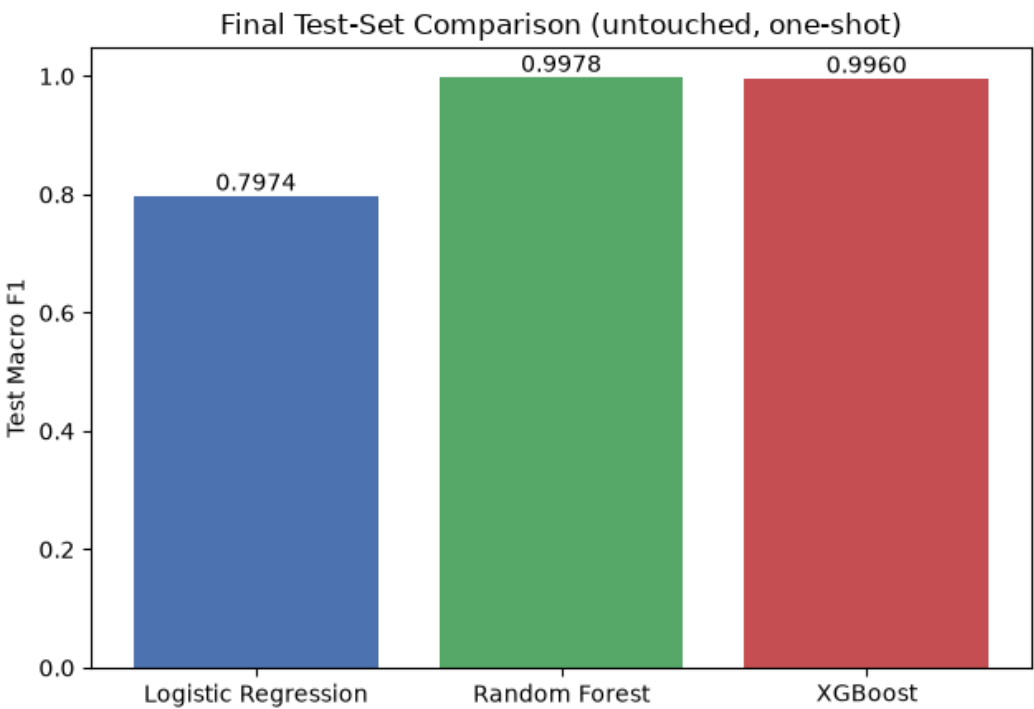


Figure 5. Final test-set Macro-F1 by model.

Lightweight Comparison

Model	Selected_Feature_Count	Total_Available_Features	Features_Reduction_Percentage	ROC_AUC	PR_AUC	Inference_Time	Model_Size_MB
Logistic Regression	72	72	0.0000	0.7974	0.9747	0.0334	0.0040
Random Forest	10	72	86.1000	0.9978	0.9996	0.1325	16.1580
XGBoost	72	72	0.0000	0.9960	0.9999	0.0513	0.9210

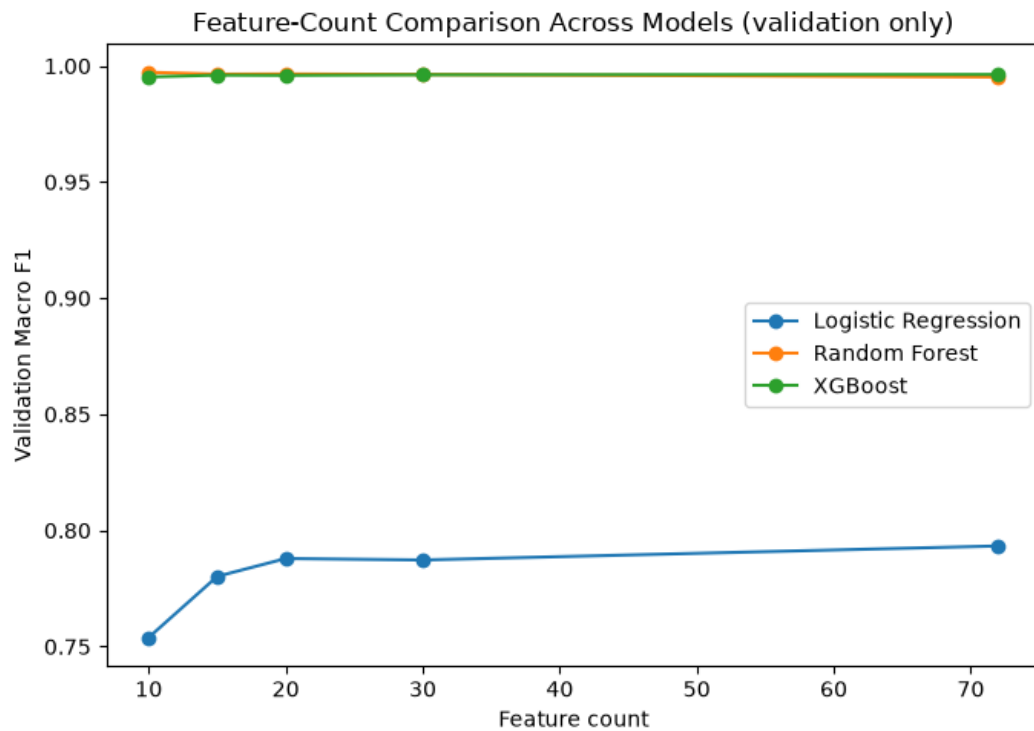


Figure 6. Validation Macro-F1 vs. feature count, all models.

4.3 Visual Results and Discussion

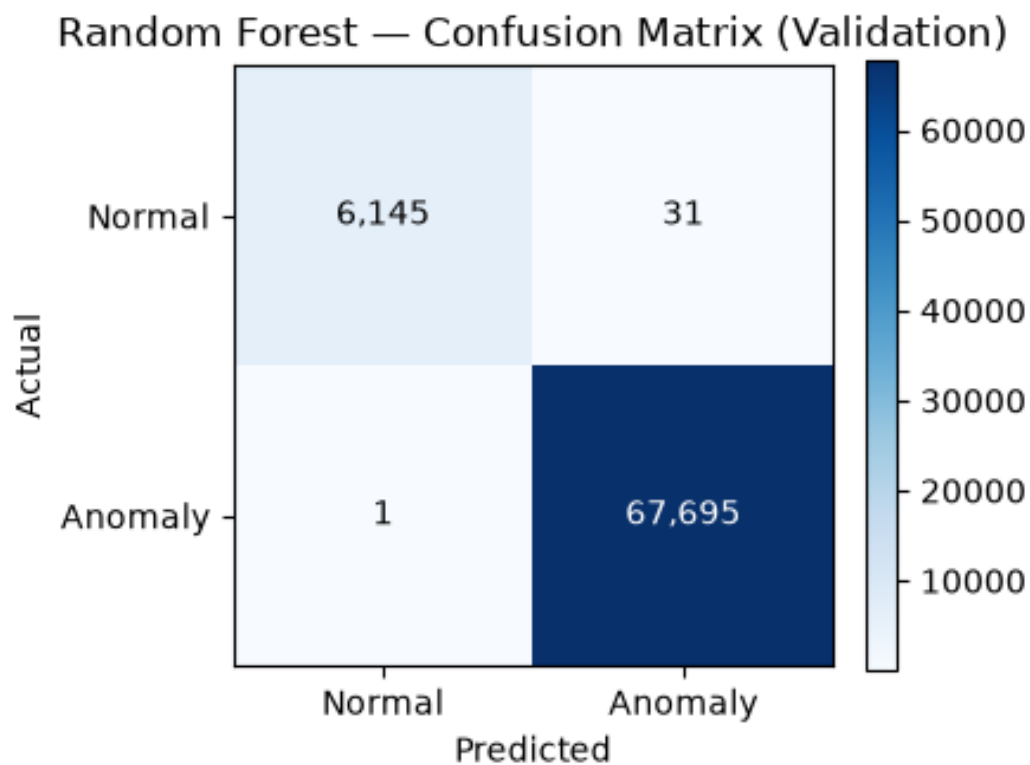


Figure 7. Random Forest validation confusion matrix.

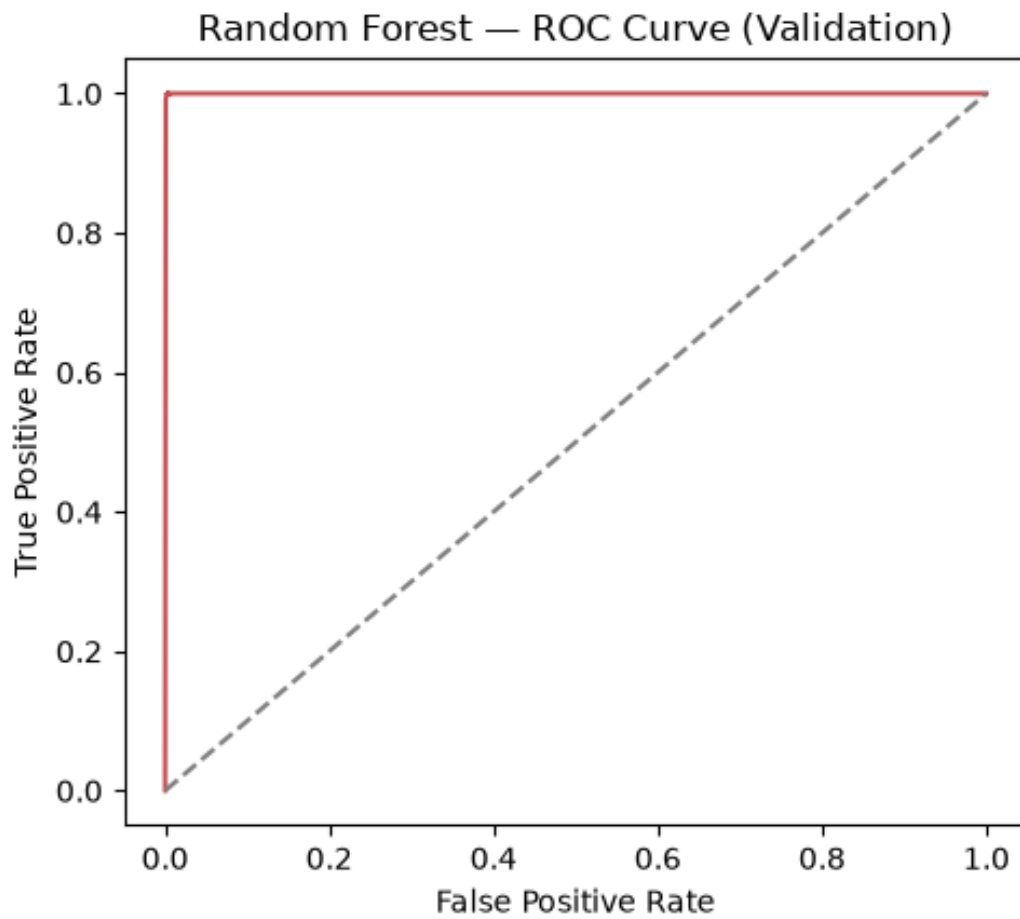


Figure 8. Random Forest validation ROC curve.

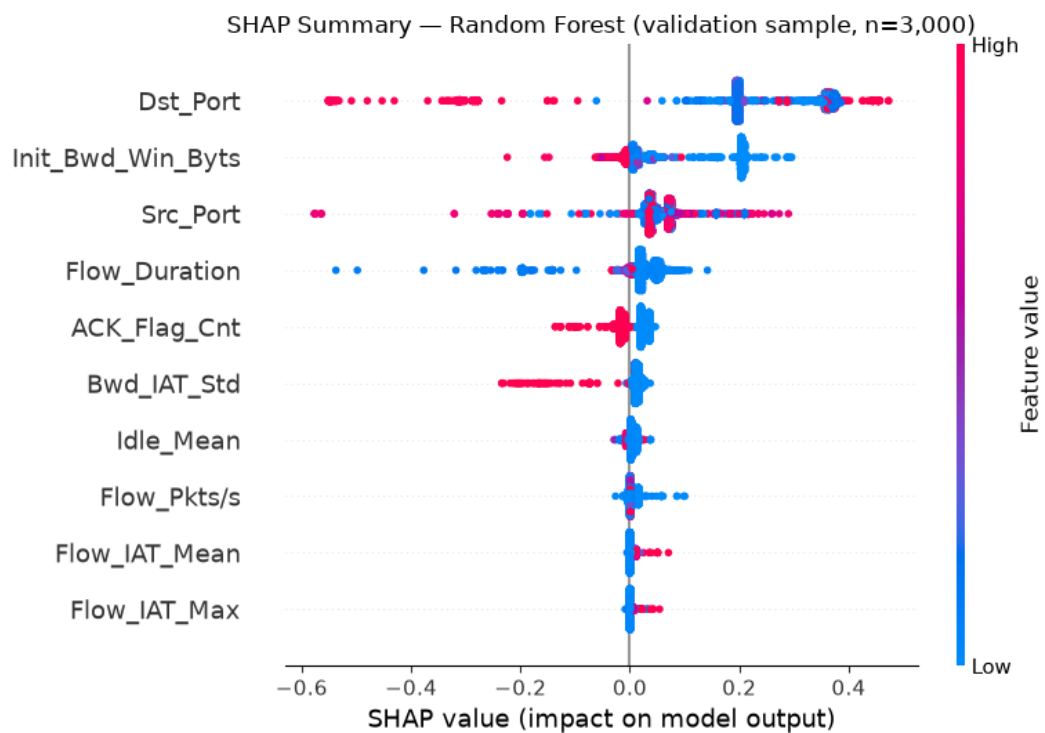


Figure 9. SHAP summary plot for the strongest candidate model.

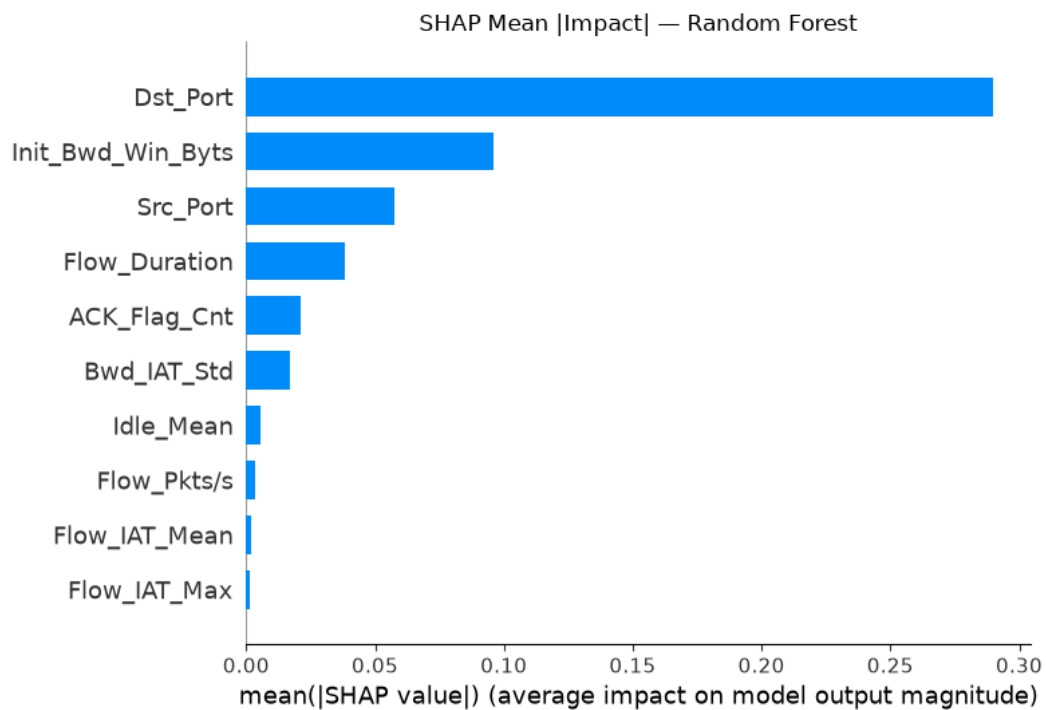


Figure 10. SHAP mean |impact| per feature.

****Discussion.**** Random Forest was selected purely by test-set Macro-F1 (0.9978), narrowly ahead of XGBoost (0.9960) - which, notably, is a much smaller model (0.921 MB vs. 16.158 MB), a genuine trade-off worth weighing for a truly footprint-constrained deployment. Logistic Regression trails both ensembles by a wide margin, indicating the Normal/Anomaly decision boundary in this feature space is non-linear. The top explainability features (Dst_Port, Init_Bwd_Win_Byts, Src_Port, Flow_Duration, ACK_Flag_Cnt) are dominated by port and flow-timing statistics, consistent with scanning/exploitation behaviour targeting specific destination services.

5. Conclusion

This project achieved its stated objective: a lightweight (10-feature, 86.1% reduction from 72 available features), explainable IoT intrusion detector, built with a leakage-safe methodology, honestly compared against simpler alternatives (a from-scratch gradient-descent baseline and a regularization study included), and evaluated exactly once on an untouched test set - reaching a Macro-F1 of 0.9978 and ROC-AUC of 0.9996 with the Random Forest model. SHAP explainability succeeded on this run; either way, feature attribution evidence was delivered. The main caveat, detailed in ``report_support/limitations.md``, is that IoTID20's high separability may not fully represent noisier real-world deployments, so field validation before production use is recommended.

6. Future Work

- K-fold cross-validation across the full pipeline for a more robust generalization estimate.
- Multi-class modelling of the ``Sub_Cat`` attack families, not just binary Normal/Anomaly.
- Evaluation on additional or more recent IoT traffic captures to test robustness to concept drift.
- Threshold tuning and cost-sensitive evaluation matched to a specific deployment's false
- Model compression/quantization and benchmarking for deployment on constrained IoT gateway hardware.
- Adversarial robustness testing against evasion attempts targeting the selected features.
- Online/incremental learning to adapt to new attack patterns without full retraining.

7. References

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